

Date: 23/06/2022 | Time: 35 minutes

1. **[4 marks]** Explain the level-of-detail rendering.
2. **[4 marks]** Consider there are two objects overlapping each other, where C_1 is the color of the foreground object and C_2 is the color of the background object.
Construct an alpha compositing formula if the foreground object has **30%** transparency and the background object is **fully opaque**.
3. **[12 marks]** Consider the following parameters for an orthographic ray-tracing:

Camera frame: $E = [-3, 7, 17]^T$, $U = [1, 0, 0]^T$, $V = [0, 1, 0]^T$, $W = [0, 0, 1]^T$

Image plane: $l = -10$, $r = 10$, $t = 15$, $b = -15$

Raster image resolution: 8×10

Sphere: $(x-5)^2 + (y-12)^2 + (z-3)^2 = 100$

Determine the ray-sphere intersection point(s) of a ray (with *length* = 25) for the upper right corner pixel of the raster image. Which intersection point will be closer to the camera?

Solution:

1. Go through the book/slides
2. $C = 0.7 C_1 + 0.3 C_2$
3. (5.75, 20.5, -2.2141634) and (5.75, 20.5, 8.2141634)

Date: 23/06/2022 | Time: 35 minutes

1. **[4 marks]** Why are triangles used in modern computer graphics?
2. **[4 marks]** Consider there are two objects overlapping each other, where $\mathbf{C}_1$ is the color of the foreground object and $\mathbf{C}_2$ is the color of the background object.
Construct an alpha compositing formula if the foreground object has **70%** transparency and the background object is **fully transparent**.
3. **[12 marks]** Consider the following parameters for an orthographic ray-tracing:

Camera frame: $\mathbf{E} = [4, 6, 15]^T$, $\mathbf{U} = [1, 0, 0]^T$, $\mathbf{V} = [0, 1, 0]^T$, $\mathbf{W} = [0, 0, 1]^T$

Image plane: $l = -15$, $r = 15$, $t = 12$, $b = -12$

Raster image resolution: 10×12

Sphere: $(x+6)^2 + (y+1)^2 + (z-5)^2 = 81$

Determine the ray-sphere intersection point(s) of a ray (with *length* = 25) for the lower left corner of the raster image. Which intersection point will be closer to the camera?

Solution:

1. Go through the book/slides
2. $\mathbf{C} = 0.3 \mathbf{C}_1 + 0.7 \mathbf{C}_2$
3. $(-9.5, -5, -2.26291952)$ and $(-9.5, -5, 12.26291952)$

1. Reflect a 2D point $P(-6, -7)$ along the diagonal AB of a square $OACB$; four vertices of the square are: $O(2,2)$, $A(2,6)$, $C(6,6)$ and $B(6,2)$. You must -

- a. **[2 marks]** Mention the steps.
- b. **[10 marks]** Determine the composite transformation matrix.
- c. **[3 marks]** Calculate and plot the reflected point P' .

2. **[5 Marks]** Write down the 2D rotation matrix that rotates by 90 degrees clockwise

Solution:

1. $P': (15.0, 14.0, 1.0)$
2. Try yourself

1. Rotate a square $OACB$ about the center of the square by 45° ; four vertices of the square are: $O(2,2)$, $A(2,10)$, $C(10,10)$ and $B(10,2)$. You must -

- a. **[2 marks]** Mention the steps to perform the task.
- b. **[10 marks]** Determine the composite transformation matrix.
- c. **[3 marks]** Find the final vertices.

2. **[5 Marks]** Write down the 4×4 3D matrix that increases the size of an object along the x and z axis by 75%.

Solution:

1. $(6, 0.34), (0.34, 6), (6, 11.66), (11.66, 6)$
2. Try yourself

1. Reflect a square OACB along a line $x = 1$; four vertices of the square are: O(3,2), A(3,6), C(7,6) and B(7,2). You must -
 - a. **[2 marks]** Mention the steps.
 - b. **[10 marks]** Determine the composite transformation matrix.
 - c. **[3 marks]** Calculate the reflected points.
2. **[5 Marks]** Write down the inverse of a 4×4 3D matrix to rotate by an angle 90° about the z-axis

Solution:

1. (-1, 2), (-1, 6), (-5, 6), (-5, 2)
2. Try yourself

1. Reflect a 2D point **P(3, 2)** along a line $y = \sqrt{3}x$. You must -
 - a. **[2 marks]** Mention the steps.
 - b. **[10 marks]** Determine the composite transformation matrix.
 - c. **[3 marks]** Calculate and plot the reflected point **P'**.
2. **[5 Marks]** Write down the 4×4 3D matrix that decreases the size of an object along the y and z axis by 30%.

Solution:

1. (0.23, 3.6)
2. Try yourself

1. Consider a viewport of size **256 x 128** in which pixel coordinates count down from the top of the image, rather than up from the bottom.

- a. **[8 Marks]** Construct the viewport matrix
- b. **[12 Marks]** Transform a 3D line AB from an orthographic view volume to the mentioned viewport. Consider the vertices of the line are A(-3, -4, -3), B(2, 4, -6) and the orthographic view volume has the following setup:

$$l = -6, r = 6, b = -7, t = 7, n = -2, f = -8$$

Solution:

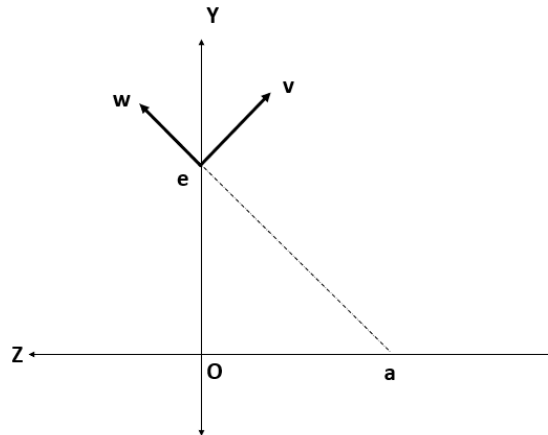
1. (a) Try yourself
(b) a'(63.5, 26.93), b'(170.17, 100.07)

1. **[8 Marks]** Derive the orthographic transformation matrix
2. **[12 Marks]** Origin **O** and basis **{z, y}** construct a 2D canonical coordinate system where **-z** is the viewing direction. Within this, a line **bc** is our model (**P_{xy}**) where vertices **b** and **c** are **(-3, 2)** and **(-6, -2)** respectively. We want to gaze at it from a new 2D camera looking upward with origin **e(-4, -6)** and basis **{w, v}**. Determine and plot the vertices **b** and **c** w.r.t camera coordinate.

Solution:

1. Try yourself
2. (-8,1), (-4, -2)

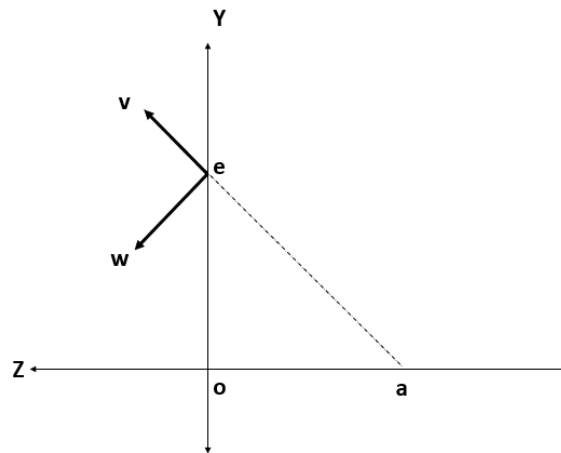
1. **[8 Marks]** Explain with the appropriate example that the frame-to-canonical transformation can be expressed as a rotation followed by a translation.
2. **[12 Marks]** Origin **O** and basis vectors **{z, y}** construct a 2D canonical coordinate system where $-z$ is the viewing direction and y is the up vector. Consider a frame coordinate with origin **e** and basis **{w, v}**. Here **e** is located on the y -axis and edge **oe** and **oa** of triangle Δoea has a length of **1 unit**. Determine the position of the point **a** w.r.t the frame coordinate.



Solution:

1. Try yourself
2. $(-1.41, 0)$

1. **[8 Marks]** In OpenGL, which form of the perspective matrix is implemented? Show that, it can be expressed in terms of field of view, aspect ratio, near and far plane.
2. **[12 Marks]** Origin **O** and basis vectors **{z, y}** construct a 2D canonical coordinate system where $-z$ is the viewing direction and y is the up vector. Consider a frame coordinate with origin **e** and basis **{w, v}**. Here **e** is located on the y -axis and edge **oe** and **oa** of the triangle Δoea has a length of **2 unit**. Determine the position of the point **a** w.r.t the frame coordinate.



Solution:

1. Try yourself
2. (0.0, -2.83)

1. [15 marks] Apply the midpoint line drawing algorithm to draw a line from (1, 1) to (3, -4) and plot the points. Show the values of the decision variables and the points for each step (in a tabular format).
2. [5 marks] Find the successive update criteria for the positive value of decision variable in Bresenham's Circle Drawing Algorithm

Solution:

1.

Octant:7				
moves x y d				

	1	1	-1	
E	1	0	3	
NE	2	-1	-3	
E	2	-2	1	
NE	3	-3	-5	
E	3	-4	-1	

2. Go through slides

1. [15 marks] Apply the midpoint algorithm to draw a circle's portions of circumference centered at (3,1) on the 4th and 5th octant with radius 6. Plot the obtained points. For each step, show values of the decision variables and the points (in a tabular format).
2. [5 marks] Find the successive update criteria for the positive value of decision variable in Bresenham's Line Drawing Algorithm

Solution:

1.

```
X Y d
0 6 -5
1 6 -2
2 6 3
3 5 0
4 4 1

octant 4

x:  [-3 -3 -3 -2 -1]
y:  [1 2 3 4 5]

octant 5

x:  [-3 -3 -3 -2 -1]
y:  [ 1  0 -1 -2 -3]
```

2. Go through slides

1. [15 marks] Apply the midpoint line drawing algorithm to draw a line from (-3, -2) to (-5, 3) and plot the points. Show the values of the decision variables and the points for each step (in a tabular format).
2. [5 marks] Briefly explain Gouraud Interpolation

Solution:

1.

Octant:3			
moves	x	y	d

	-3	-2	-1
E	-3	-1	3
NE	-4	0	-3
E	-4	1	1
NE	-5	2	-5
E	-5	3	-1

2. Go through slides/book

1. [15 marks] Apply the midpoint algorithm to draw a circle's portions of circumference centered at (-5,-3) on the 3rd and 7th octant with radius 7. Plot the obtained points. For each step, show the values of the decision variables and the points (in a tabular format).
2. [5 marks] What is graphics pipeline? State the difference between hardware and software pipeline.

Solution:

1.

```
x y d
0 7 -6
1 7 -3
2 7 2
3 6 -3
4 6 6
5 5 7

octant 3
x:  [-5 -6 -7 -8 -9 -10]
y:  [4 4 4 3 3 2]

octant 7
x:  [-5 -4 -3 -2 -1 0]
y:  [-10 -10 -10 -9 -9 -8]
```

2. Go through slides/book